Year 11 Mathematics

Handling Data 5

9.1 Frequency Tables

Frequency Tables

STARTER · 5 MIN

Date: _____ **Title:** Frequency Tables

LEARNING OBJECTIVES

- Find the mean, median and mode from a frequency table
- Estimate the mean from a grouped frequency table
- Find the modal class and the class containing the median

STARTER *Answer in your book. No calculators.*

1. Find the mean of 2, 4, 6, 8
2. Find the midpoint of 10 and 20
3. Work out $3 \times 5 + 4 \times 15$
4. Find the median of 1, 3, 3, 7, 9, 10

Starter — Answers

STARTER

Tick or correct in green pen

1. $20 \div 4 = \mathbf{5}$
2. $(10 + 20) \div 2 = \mathbf{15}$
3. $15 + 60 = \mathbf{75}$
4. $(3 + 7) \div 2 = \mathbf{5}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the mean, median and mode from a frequency table	☐
SILVER	I can estimate the mean from a grouped frequency table	☐
GOLD	I can find the modal class and the class containing the median	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Frequency Tables

INTRODUCTION · 3 MIN

A frequency table shows how many times each value or group occurs.

Mean

$\Sigma fx \div \Sigma f$

Add an f × x column

Midpoint

Use the midpoint of each group.

$10 \leq x < 20 \rightarrow 15$

Modal class

The group with the highest frequency.

Write the whole group

Median

Find the $(n + 1) \div 2$ th value.

$n = 30 \rightarrow$ 15.5th value

Examples 1 and 2 — Averages from a Frequency Table

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint mean = $\frac{\Sigma fx}{\Sigma f}$

Example 1

Find the mean number of goals.

Goals	Frequency
0	4
1	6
2	3
3	2

Example 2

Find the median and the mode.

Goals	Frequency
0	4
1	6
2	3
3	2

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

fx: 0, 6, 6, 6

Σfx = 18

Σf = 15

Mean = 18 ÷ 15

= 1.2

Tip: divide by the total frequency.

Example 2

15 values: the 8th value

Running totals 4, 10

Median = 1

Mode **= 1**

Tip: add the frequencies as you go.

Task 1

PRACTICE · 9 MIN

BRONZE *Use the table of pets*

Hint mean = $\frac{\Sigma fx}{\Sigma f}$

1. How many people were asked?
2. Write down the mode
3. Find the range
4. Find Σfx
5. Find the mean

★ Find the median

Pets	Frequency
0	7
1	9
2	5
3	3
4	1

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $7 + 9 + 5 + 3 + 1 = \mathbf{25}$
2. **1 pet**
3. $4 - 0 = \mathbf{4}$
4. $0 + 9 + 10 + 9 + 4 = \mathbf{32}$
5. $32 \div 25 = \mathbf{1.28}$

★ 13th value, **1 pet**

Examples 3 and 4 — Estimating the Mean

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint mean ≈ $\frac{\Sigma fx}{\Sigma f}$ x = midpoint

Example 3

Estimate the mean time.

Time t (s)	Frequency
$0 \leq t < 10$	5
$10 \leq t < 20$	8
$20 \leq t < 30$	7

Example 4

Why is the mean from a grouped table only an estimate?

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Midpoints 5, 15, 25

$\Sigma fx = 25 + 120 + 175$

$= 320$

Mean $\approx 320 \div 20$ **$= 16$ s**

Tip: multiply each midpoint by its frequency.

Example 4

We use the midpoints

The exact values are not known

Tip: the data is grouped.

Task 2

PRACTICE · 9 MIN

SILVER *Use the table of masses*

Hint mean ≈ $\frac{\Sigma fx}{\Sigma f}$ x = midpoint

1. Find the total frequency
2. Write down the modal class
3. Write down the midpoints
4. Find Σfx
5. Estimate the mean

Mass m (kg)	Frequency
$40 \leq m < 50$	4
$50 \leq m < 60$	10
$60 \leq m < 70$	12
$70 \leq m < 80$	4

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $4 + 10 + 12 + 4 = \mathbf{30}$

2. $\mathbf{60 \leq m < 70}$

3. **45**, **55**, **65**, **75**

4. $180 + 550 + 780 + 300 = \mathbf{1810}$

5. $1810 \div 30 = \mathbf{60.3\ kg}$

Examples 5 and 6 — Median Class and Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint median: the $\frac{n+1}{2}$ th value

Example 5

(2 marks)

Which class contains the median?

Time t (s)	Frequency
$0 \le t < 10$	5
$10 \le t < 20$	8
$20 \le t < 30$	7

Example 6

(3 marks)

Groups 1–5, 6–10 and 11–15 have frequencies 6, k and 2. The estimated mean is 7. Find k.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

20 values: the 10.5th value

Running totals 5, 13

$10 \leq t < 20$

Tip: find where the 10.5th value lies.

Example 6

Midpoints 3, 8, 13

$18 + 8k + 26 = 7(8 + k)$

$44 + 8k = 56 + 7k$

$k = 12$

Tip: Σfx = mean × Σf.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint median: the $\frac{n+1}{2}$ th value

1. Estimate the mean time (4 marks)

2. Which class contains the median? (1 mark)

3. Estimate the percentage who took 15 minutes or more (2 marks)

4. Explain why the range cannot be found exactly (1 mark)

5. Another 20 people had a mean of 12 minutes. Find the mean of all 50. (2 marks)

Time t (min)	Frequency
$0 \le t < 5$	6
$5 \le t < 10$	11
$10 \le t < 15$	8
$15 \le t < 20$	5

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $285 \div 30 =$ **9.5 minutes**

2. 15.5th value, **$5 \leq t < 10$**

3. $5 \div 30 \times 100 =$ **16.7%**

4. **The exact values are not known**

5. $285 + 240 = 525$, $525 \div 50 =$ **10.5 minutes**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Values 1, 2, 3 have frequencies 2, 5, 3. Find the mean.

SILVER Write down the midpoint of $20 \leq x < 30$

GOLD n = 40. Which value is the median?

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $(2 + 10 + 9) \div 10 =$ **2.1**

SILVER $(20 + 30) \div 2 =$ **25**

GOLD $(40 + 1) \div 2 =$ **20.5th value**

Exam Practice — Averages from a Frequency Table

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint mean = $\frac{\Sigma fx}{\Sigma f}$

Question 1

(3 marks)

Books read by 25 students:

Books	0	1	2	3	4
Frequency	3	7	8	5	2

(a) Write down the mode. (1)

.....

(b) Work out the mean. (2)

.....

Question 2

(2 marks)

Shoe sizes of 24 students:

Find the median shoe size.

Size	4	5	6	7	8
Frequency	2	6	9	4	3

.....

Exam Practice — Estimating the Mean

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint mean ≈ $\frac{\Sigma fx}{\Sigma f}$ x = midpoint

Question 3

(4 marks)

The table shows the times of 20 people.

Estimate the mean time.

Time (t minutes)	Frequency
$0 < t \leq 10$	4
$10 < t \leq 20$	9
$20 < t \leq 30$	5
$30 < t \leq 40$	2

..... minutes

Question 4

(2 marks)

Use the table in Question 3.

(a) Write down the modal class. (1)

.....

(b) Why is the mean only an estimate? (1)

.....

Exam Practice — Median Class and Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint median: the $\frac{n+1}{2}$ th value

Question 5

(3 marks)

Heights of 40 students:

Height (h cm)	Frequency
$140 < h \leq 150$	5
$150 < h \leq 160$	12
$160 < h \leq 170$	15
$170 < h \leq 180$	8

(a) Which class has the median? (1)

.....

(b) What % are taller than 160 cm? (2)

..... %

Question 6

(4 marks)

Groups 1–5, 6–10 and 11–15 have frequencies 4, k and 6.

The estimated mean is 8.5

Find k.

k = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) 2 — B1

(b) 0 + 7 + 16 + 15 + 8 = 46 — M1

46 ÷ 25 = 1.84 — A1

Tip: multiply each value by its frequency.

Question 2

(24 + 1) ÷ 2 = 12.5th value — M1

running totals 2, 8, 17

size 6 — A1

Tip: add the frequencies until you pass 12.5

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

midpoints 5, 15, 25, 35 — M1

20 + 135 + 125 + 70 = 350 — M1

350 ÷ 20 — M1

= 17.5 minutes — A1

Tip: divide by 20 people, not 4 groups.

Question 4

(a) 10 < t ≤ 20 — B1

(b) midpoints are used, — B1

not the real times

Tip: we do not know the exact times.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

(a) 20.5th value

160 < h ≤ 170 B1

(b) 15 + 8 = 23 M1

23 ÷ 40 × 100 = 57.5% A1

Tip: running totals 5, 17, 32.

Question 6

midpoints 3, 8, 13 M1

12 + 8k + 78 = 8.5(10 + k) M1

90 + 8k = 85 + 8.5k M1

k = 10 A1

Tip: Σfx = mean × Σf.